



Liquid Culture Preparation

Designing Life with AI

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Description

This method is used to grow bacteria in a nutrient-rich liquid medium, allowing them to multiply and reach a high density for further experiments.

Material

- LB medium (or other appropriate liquid growth medium)
- Antibiotic stock solution (if strain carries resistance cassette)
- Bacteria (from plate or glycerol stock)

Steps

1. Pick a (single) colony and inoculate into x **mL** of LB (or another appropriate medium) in a sterile tube or flask. Do not fill the vessel more than $\sim 20\text{--}30\%$ of its nominal volume (better aeration).
2. If the strain carries an antibiotic resistance cassette, add x **μL** of antibiotic from stock to reach a 1:1000 antibiotic/medium ratio.
3. Incubate in a shaking incubator at the strain-appropriate temperature (37°C for *E. coli*, and 30°C for *P. putida*) and ~ 200 rpm until desired growth (typically overnight $\approx 16\text{--}18$ h).

Growth can be known with an OD_{600} measurement.